VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE
Near Ramoji Film City, Deshmukhi Village, Pochampally Mandal, Yadadri Bhuvanagiri Dist.
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DIGITAL INDIA INNOVATION VIGNAN'S INNOVATIVE PROJECT

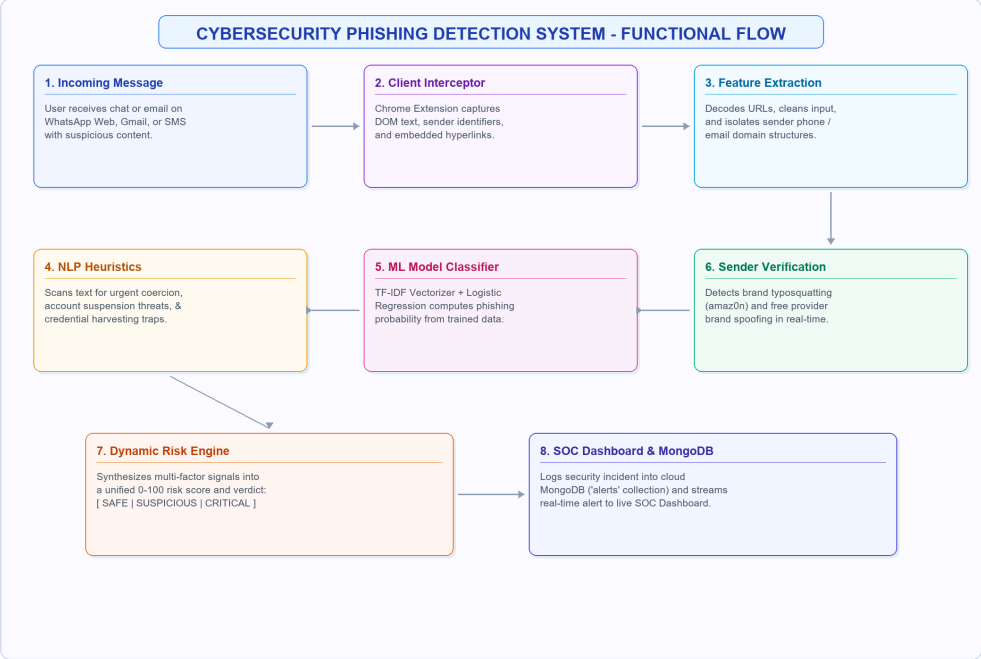
VIKSIT BHARATH 2047

YEAR & SEMESTER: III Year & II Sem **ACADEMIC YEAR: 2026-27**

ABSTRACT

Batch No.	**3B2**
Title of the Project	**Cybersecurity Phishing Detection System**
Student Details	**Roll No** / **Name of the Student** 24891A0594 KOTHAPALLY BINDUSREE 24891A0596 MALLEPAKULA BHUVANISHWARI 24891A0575 DUNGROTHU NIKHIL 24891A0573 DEVARASHETTY RAM NARAYAN
Guide Name	**Mr. VSSPLN Balaji Lanka**
Name of the Subject	Cyber Security, AI&ML
Abstract	Phishing attacks have rapidly evolved from traditional deceptive emails into sophisticated, multi-channel social engineering campaigns spanning modern messaging applications such as WhatsApp Web, instant messaging platforms, SMS, and corporate emails. Conventional security defense mechanisms relying strictly on static domain blocklists or single-vector heuristics are consistently bypassed by zero-day typosquatting domains, brand impersonation using free email services, and fast-flux URL redirects. Cybersecurity Phishing Detection System is an end-to-end, multi-vector defense solution engineered to intercept and neutralize phishing threats in real time across multiple communication channels. The solution employs a lightweight browser extension and client-side listeners that capture incoming textual payloads, embedded URLs, and sender identifiers directly from active web sessions without disrupting user experience. The intercepted data is processed by a hybrid AI risk evaluation engine combining Natural Language Processing (NLP) heuristics, a TF-IDF with Logistic Regression machine learning classifier, and deep sender reputation analytics. The system evaluates sender authenticity by detecting brand typosquatting via Levenshtein-distance heuristics, flagging unauthorized free-provider impersonations (e.g., claiming to be PayPal or Amazon from a Gmail address), and assessing international phone prefixes. A multi-factor risk scoring algorithm synthesizes text urgency, domain legitimacy, and sender integrity into an actionable threat score (0?100) and categorized verdict (SAFE, SUSPICIOUS, HIGH RISK, CRITICAL). Confirmed security incidents are logged to a cloud-hosted MongoDB alerts database and streamed in real time to a centralized SOC Web Dashboard, providing security analysts with instant situational awareness and automated incident mitigation.

Image/ Functional Diagram



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